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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Made in UTM
Project / Solution Name	Laluan: a career GPS for the AI era
Name (Leader)	Tan Hong Sheng
Email & Phone (Leader)	tanhongsheng050204@gmail.com +60 11-1691 8805
University Affiliation(s)	Universiti Teknologi Malaysia (UTM)
Live Prototype URL	https://laluan-psi.vercel.app (API docs: https://laluan-api.onrender.com/docs)
Code Repository URL (public / access granted to judges)	https://github.com/tanhongsheng050204-code/SDC-HACKATHON-2026-XMUM---Made-In-UTM (Public)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The user opens a React single-page app served by **Vercel**. It calls a **FastAPI** backend over HTTPS, running as a **Docker** container on **Render** (Singapore). Sign-in is optional (Google login through **Supabase Auth**); profiles, progress, Future Cards and trend snapshots are kept in **Supabase Postgres** with row-level security.

When a CV arrives, the backend reads it in memory (our own OCR for scans) and removes names, emails, phone and IC numbers before any model sees it. An open model, **Qwen2.5-7B on RunPod Serverless**, reads skills from the scrubbed text; **Google Gemini** writes coach answers, proof-check questions and summaries, and Gemini embeddings match unfamiliar phrases. Without keys, rules and templates take over.

Matching, scenarios, forecasts and pathways are deterministic Python over versioned JSON reference data (WEF, O*NET, courses, SDG map). AI-written text passes a grounding check against these sources before it is shown.

A scheduled trend pipeline collects research, job-ad, Malaysian statistics and developer-activity signals, forecasts each to 2030 with 80% ranges and a backtest, and holds the new snapshot until an admin approves it. **GitHub Actions** runs the tests and build on every push. Diagram: Appendix.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.6.3; Python 3.12; CSS; SQL	Languages	Apache-2.0 / PSF / – / –
React 18.3.1, React DOM 18.3.1	User interface	MIT
Vite 5.4.11, @vitejs/plugin-react 4.3.4	Frontend build and dev server	MIT
@supabase/supabase-js 2.45.4	Google sign-in from the browser	MIT
qrcode 1.5.4	QR code on the shareable Future Card	MIT

Name & Version	Purpose in Project	License
@fontsource-variable/inter 5.3.0	Self-hosted Inter typeface	SIL OFL 1.1
@sentry/react 8.38.0, sentry-sdk 2.x	Error tracking (frontend and API, optional)	MIT
FastAPI 0.152.0 (Starlette 1.9.0)	HTTP API	MIT / BSD-3
Pydantic 2.14.1	Request validation (also rejects demographic fields)	MIT
Uvicorn 0.54.0	ASGI server	BSD-3
google-genai 1.75.0	Gemini API client	Apache-2.0
HTTPX 0.28.1	Calls to RunPod and the trend data APIs	BSD-3
psycopg 3.3.6 (with pool)	Postgres driver for Supabase	LGPL-3.0
PyJWT 2.15.1	Verifying Supabase login tokens	MIT
pypdf 6.21.1, python-multipart 0.0.32	Reading uploaded PDF CVs in memory; file uploads	BSD-3 / Apache-2.0
RapidOCR 2.1.0, ONNX Runtime 1.31.0, pypdfium2 5.14.0, Pillow 13.0.0	Reading scanned CVs and photos with our own OCR (RunPod worker first, else the API server), before scrubbing	Apache-2.0 / MIT / Apache-2.0 or BSD-3 / MIT-CMU
RunPod worker templates: vLLM, faster-whisper	Serving Qwen2.5-7B and Whisper on RunPod Serverless	Apache-2.0 / MIT
pytest 8.4.2, Vitest 2.1.8, Testing Library, ESLint 9.15.0, axe-core 4.10.2	Tests, lint and accessibility checks (development only)	MIT / MIT / MIT / MIT / MPL-2.0
PyMuPDF	One-off script reading WEF Figure 2.3 from the report PDF; not part of the app	AGPL-3.0
Docker images: python:3.12-slim, node:20-alpine, nginx:1.27-alpine	API container and local docker compose run	PSF / MIT / BSD-2

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API: gemini-3.8-flash, gemini-embedding-001, Google Search grounding	Coach answers, proof-check questions, interview feedback, summaries, survey reading; embeddings for unfamiliar skill phrases; finding a real source for each emerging-role prediction. The app runs without it (rules and templates).	Paid tier (team's own key)
RunPod Serverless API	Qwen2.5-7B reads skills from scrubbed CV text; Whisper turns voice into text; OCR worker reads scanned CVs	Sponsor-provided credit
Supabase Auth API	Optional Google sign-in	Free
arXiv API (Cornell University)	AI research paper counts per capability (trend signal)	Free
OpenAlex API (OurResearch)	Fallback paper counts; papers from Malaysian institutions	Free
Hacker News Search API (Algolia)	Skill mentions in “Who is hiring?” threads	Free
freehire.me public API	Skill mentions in Malaysian job postings; newest openings per role, linked to the original posts	Free, no key
OpenDOSM / data.gov.my API (Department of Statistics Malaysia)	Employment by occupation, monthly labour force, employment by sector, consumer prices	Free, no key
Stack Exchange API, GitHub REST API, npm downloads API, pypistats.org	Developer attention per skill (questions, new repositories, package downloads)	Free, no key
Gmail SMTP (team account, app password)	Optional weekly digest and job-alert emails	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Vercel	Hosts the built React frontend (deployed from GitHub)	Hobby (free), global edge
Render	Runs the FastAPI backend as a Docker container (render.yaml + backend/Dockerfile)	Free web service, Singapore
Supabase	Postgres database (row-level security on every table) and Auth (Google login)	Free plan, Singapore
RunPod Serverless	GPU endpoints for Qwen2.5-7B (vLLM), faster-whisper and our OCR worker; scale to zero when idle	Pay per second, from the SDC sponsor credit
GitHub + GitHub Actions	Public source repository; tests on SQLite and Postgres and the frontend build on every push; scheduled trend-refresh trigger	Free, public repository
Sentry	Error tracking for the frontend and API (optional)	Free plan
Docker / Docker Compose	One-command local run of the whole app (docker compose up --build)	Local

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini 3.8 Flash (gemini-3.8-flash)	Google, via the Gemini API	Pre-trained, called by API	Gemini API Terms of Service
Gemini Embedding (gemini-embedding-001)	Google, via the Gemini API	Pre-trained, called by API	Gemini API Terms of Service
Qwen2.5-7B-Instruct	Alibaba Qwen team (Hugging Face); served with vLLM on our RunPod endpoint	Pre-trained, self-hosted inference	Apache-2.0
Whisper small (faster-whisper)	OpenAI weights, SYSTRAN faster-whisper; on our RunPod endpoint	Pre-trained, self-hosted inference	MIT
PaddleOCR PP-OCRv4 English recogniser + RapidOCR text detector (ONNX)	PaddlePaddle / RapidAI; run on our RunPod worker or the API server	Pre-trained, run locally	Apache-2.0

No model is trained or fine-tuned by the team. The model in use is shown in the app. AI never ranks roles, calculates forecasts or chooses courses: those are deterministic code over cited data.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
WEF Future of Jobs Report 2025, Fig. 2.3 (net job growth 2025–2030)	reports.weforum.org/docs/WEF_Future_of_Jobs_Report_2025.pdf	Public	WEF terms of use; cited on every role	N
O*NET 30.0 task statements	onetcenter.org	Public	CC BY 4.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
OpenDOSM: wages_occupation, lfs_month, employment_sector, cpi_headline; DOSM Salaries & Wages 2025	open.dosm.gov.my, api.data.gov.my	Public	CC BY 4.0	N
TalentCorp MyCOL 2024/2025	talentcorp.com.my/mycol	Public report (matched by job title, linked)	Cited with link	N
arXiv and OpenAlex paper counts	export.arxiv.org/api, api.openalex.org	Public (counts only)	arXiv API terms; CC0	N
Hacker News “Who is hiring?” mention counts	hn.algolia.com/api	Public (counts only, no posts stored)	HN / Algolia API terms	N
Malaysian job postings (mention counts; live openings)	freehire.me/docs/api	Public (counts stored; postings shown live, not stored)	freehire terms; project MIT	N
Stack Overflow, GitHub, npm and PyPI activity counts	api.stackexchange.com, api.github.com, api.npmjs.org, pypistats.org	Public (counts only)	Each API's terms	N
Anthropic Economic Index	huggingface.co/datasets/Anthropic/EconomicIndex	Public (aggregates)	CC BY	N
OpenAI “GPTs are GPTs” exposure scores	github.com/openai/GPTs-are-GPTs	Public	MIT	N
ILO Working Paper 140, Annex Table A1	ilo.org	Public	CC BY 4.0	N
UN SDG goals, targets and colours	sdgs.un.org/goals	Public	UN SDG guidelines	N
Course catalogue (prices, hours, HRD Corp status)	Providers' pages, dated in backend/data/courses.json	Self-collected	Facts with links to providers	N
Roles, skills and SDG map	backend/data/	Self-created	Apache-2.0 (this repository)	N
Demo personas, evaluation CVs, sample company (190 staff) and sample class (52 students)	backend/data/	Synthetic (all invented)	Apache-2.0 (this repository)	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Landing-page photos (3)	Unsplash: Gema Saputera, Mimi Thian, Vitaly Gariev	Public	Unsplash License	N (people shown, not linked to any data)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic), Claude Opus models	Planning documents; writing and refactoring backend and frontend code and tests; screen layouts (modelled on a reference site the team chose); Docker and deployment files; extracting WEF figures; drafting Bahasa Melayu and Chinese interface text; drafting these declarations	Major
Google Gemini (inside the product)	Drafting automate / assist / human labels for O*NET tasks, for later human review	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team chose the problem, users, features, design reference and data sources, and directed every change. AI-written code ships only with tests: about 490 automated backend tests, frontend component tests, a type-checked build and lint, run in CI on SQLite and Postgres for every push. Each feature was clicked through in a browser, including at phone width. Problems found and their fixes are logged in REVIEW_LOG.md; design choices in DECISIONS.md. In the app, AI never acts alone: the coach only suggests, trend updates need an admin's approval, and AI text is checked against the catalogue.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
RunPod	Serverless GPU platform and credits	Hosts our three AI workers: Qwen2.5-7B-Instruct (vLLM) reads skills from scrubbed CV text, faster-whisper transcribes voice input, and an OCR worker reads scanned CVs. Raw CV text and audio stay on open models we control instead of a third-party AI API.
Replit	Platform	Used as a validation sandbox to test the web application in a clean cloud environment, verify dependency installation, and validate runtime behavior outside of local machines.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Important but replaceable. RunPod carries our privacy design: CVs and voice are read by open models we run, not a third-party API. Every RunPod feature has a fallback: OCR also runs on the API server, CV reading falls back to Gemini on scrubbed text or to dictionary matching, and voice is optional (typing works). With no AI keys at all, the app still works with rules and templates.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Collected: confirmed skills, current role, years of experience, chosen SDG goals and course progress, stored in Supabase Postgres (row-level security) under a random profile ID. Sign-in is optional (Google, via Supabase Auth). The API rejects age, gender and background fields.
CVs, chats, voice and interview answers: processed in memory, never stored. Names, emails, phone and Malaysian IC numbers are removed before any text goes to RunPod or Gemini, and the user sees what was removed and what is sent.
Third parties: only scrubbed text and questions about a person's own results reach the AI APIs.
Retention: users can download or delete everything from Settings; profiles not saved to a Google account are deleted after 30 days.
HR and class views: files with name, email, IC, phone or staff-ID columns are refused; groups under three are hidden.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hallucination: AI text could invent courses or numbers. Courses, roles and figures come only from reference data, and AI sentences are checked against those sources and flagged if unsupported.
Forecast uncertainty: WEF figures are global employer estimates and trend signals are noisy. Three scenarios, 80% ranges, shrinkage towards no change and a published backtest (direction 14 of 19; sizes rougher).
Noisy or tech-heavy signals: capped adjustments, minimum mention counts, and a person approves each update.
Bias: no demographic inputs; a public fairness check and tests confirm names do not change rankings.
Provider dependence and rate limits: arXiv falls back to OpenAlex; every AI feature has an offline path; the last approved snapshot stays live.
Free hosting: the API sleeps when idle (about a minute to wake); the app shows a notice and retries.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature: Tan Hong Sheng (Team Leader)

Date: 24 October 2026

Appendix: System Block Diagram

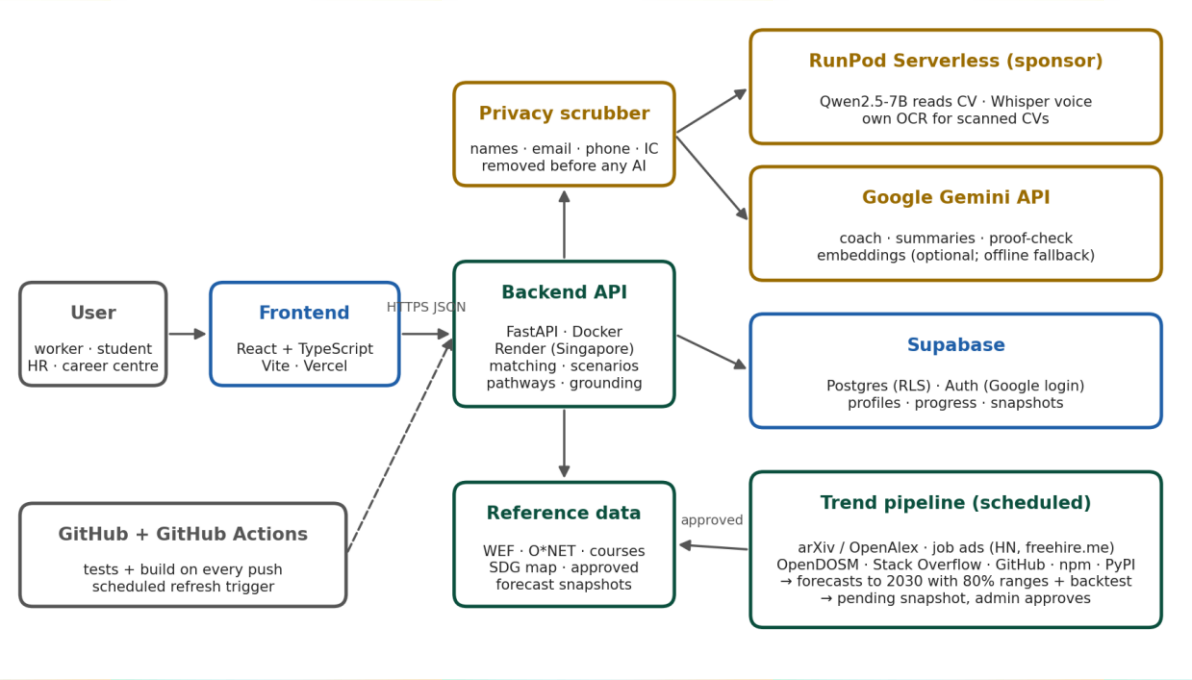


Figure A1. Laluan system architecture. Amber: AI services (behind the privacy scrubber); green: our own code and data; blue: hosting and database.